Chapter 2: Natural Resources: Land, Water, Air and Conservation

Class 7 - Social Science | Unit: Our Environment (Geography)

CHAPTER OVERVIEW & CORE FOUNDATIONAL CONCEPTS

Investigates Earth ecological domains - lithosphere, hydrosphere, atmosphere, and biosphere - and sustainable resource management.

KEY LEARNING OBJECTIVES

1. Examine the components of natural, human, and human-made environments.
2. Understand the hydrological cycle and global freshwater distribution.
3. Evaluate factors contributing to environmental degradation and practical conservation strategies.

1. EARTH SPHERES AND ECOSYSTEM BALANCE

The lithosphere is the solid crust of rocks and minerals. The hydrosphere encompasses water bodies. The atmosphere is the thin protective gaseous blanket held by gravity. The narrow zone where life exists is the biosphere.

2. FRESHWATER SCARCITY AND CONSERVATION MEASURES

While Earth is 71 percent water, less than 3 percent is fresh, with most locked in polar glaciers. Conservation practices include rainwater harvesting, drip irrigation in agriculture, and industrial effluent treatment.

DIAGNOSTIC FOCUS & COMMON MISCONCEPTIONS

Students often overlook that freshwater distribution is finite and assume surface water runoff replenishment is instantaneous regardless of deforestation and soil erosion.

PREREQUISITE REINFORCEMENT QUESTIONS

Q1: Define an ecosystem and give one example of human intervention disrupting it.

Expected Answer: An ecosystem is a community of living organisms interacting with their physical environment; deforestation for urban expansion disrupts nutrient and rainfall cycles.

Q2: Explain the working principle of rainwater harvesting.

Expected Answer: Collecting and filtering rainwater runoff from rooftops and open surfaces into storage tanks or borewells to recharge groundwater aquifers.